77. *(b)* $\cot(\theta+\phi) = \dfrac{1}{\tan(\theta+\phi)} = \dfrac{1-\tan\theta\tan\phi}{\tan\theta+\tan\phi}$

$$= \frac{1}{\tan\theta+\tan\phi} - \frac{\tan\theta\tan\phi}{\tan\theta+\tan\phi}$$

$$= \frac{1}{\tan\theta+\tan\phi} - \frac{1}{\left(\dfrac{1}{\tan\theta}+\dfrac{1}{\tan\theta}\right)}$$

$$= \frac{1}{\tan\theta+\tan\phi} - \frac{1}{(\cot\theta+\cot\phi)} = \frac{1}{a}-\frac{1}{b}$$

[$\because \tan\theta+\tan\phi = a$, or $\cot\theta+\cot\phi = b$]

78. *(b)* $\because \tan 75° = \tan(45°+30°)$

$\Rightarrow \tan 75° = \dfrac{\tan 45° + \tan 30°}{1-\tan 45°\tan 30°} \Rightarrow \tan 75° = \dfrac{1+\tan 30°}{1-\tan 30°}$

$\Rightarrow \tan 75° - \tan 75°\cdot\tan 30° = 1+\tan 30°$

$\Rightarrow \tan 75° - \tan 30° - \tan 75°\cdot\tan 30° = 1$

79. *(b)* $\sin^2(15°+A) - \sin^2(15°-A) = \sin(15°+A+15°-A)$
$\cdot\sin(15°+A-15°+A)$

[$\because \sin^2 A - \sin^2 B = \sin(A+B)\cdot\sin(A-B)$]

$= \sin 30°\sin(2A) = \dfrac{1}{2}\sin 2A$

80. *(a)* $\dfrac{\sin 38° - \cos 68°}{\cos 68° + \sin 38°} = \dfrac{\cos(90°-38°)-\cos 68°}{\cos 68° + \sin(90°-52°)}$

$$= \frac{\cos 52° - \cos 68°}{\cos 68° + \cos 52°} = \frac{2\sin\left(\dfrac{52°+68°}{2}\right)\sin\left(\dfrac{68°-52°}{2}\right)}{2\cos\left(\dfrac{52°+68°}{2}\right)\cos\left(\dfrac{68°-52°}{2}\right)}$$

$$= \frac{\sin 60°\cdot\sin 8°}{\cos 60°\cdot\cos 8°} = \tan 60°\cdot\tan 8° = \sqrt{3}\tan 8°$$

81. *(c)* $2\cos x - (\cos 3x + \cos 5x)$

$= 2\cos x - 2\cos 4x\cdot\cos x = 2\cos x(1-\cos 4x)$

$= 2\cos x\cdot 2\sin^2 2x = 4\cos x(2\sin x\cos x)^2 = 16\cos^3 x\sin^2 x$

82. *(c)* $\cos 4x = 1 - 2\sin^2 2x = 1 - 2(2\sin x\cos x)^2$

$= 1 - 2(4\sin^2 x\cos^2 x) = 1 - 8\sin^2 x\cos^2 x$

83. *(a)* $2\sin A\cos^3 A - 2\sin^3 A\cos A$

$= 2\sin A\cos A(\cos^2 A - \sin^2 A)$

$= \sin 2A(\cos 2A) = \dfrac{2\sin 2A\cos 2A}{2} = \dfrac{1}{2}\sin 4A$

84. *(c)* Given, $a\cos\theta - b\sin\theta = c$

On squaring both sides, we get

$a^2\cos^2\theta + b^2\sin^2\theta - 2ab\cos\theta\sin\theta = c^2$

$\Rightarrow a^2(1-\sin^2\theta) + b^2(1-\cos^2\theta) - 2ab\sin\theta\cos\theta = c^2$

$\Rightarrow a^2+b^2-c^2 = a^2\sin^2\theta + b^2\cos^2\theta + 2ab\sin\theta\cos\theta$

$\Rightarrow (a\sin\theta + b\cos\theta)^2 = a^2+b^2-c^2$

$\Rightarrow a\sin\theta + b\cos\theta = \pm\sqrt{a^2+b^2-c^2}$

85. *(d)* Given, $\cos x + \cos^2 x = 1 \Rightarrow \cos x = 1-\cos^2 x = \sin^2 x$ ...(i)

$\sin^8 x + 2\sin^6 x + \sin^4 x = (\cos x)^4 + 2\cos^3 x + \cos^2 x$

$= (\cos^2 x + \cos x)^2$ [from Eq. (i)]

$= (1)^2 = 1$

86. *(b)* $\dfrac{n}{m} = \dfrac{\sec x - \cos x}{\operatorname{cosec} x - \sin x} = \dfrac{\dfrac{1-\cos^2 x}{\cos x}}{\dfrac{1-\sin^2 x}{\sin x}}$

$$= \frac{\sin^2 x}{\cos x}\cdot\frac{\sin x}{\cos^2 x} = \left(\frac{\sin x}{\cos x}\right)^3$$

$\Rightarrow (\tan x)^3 = \left(\dfrac{n}{m}\right) \Rightarrow \tan x = \left(\dfrac{n}{m}\right)^{1/3}$

87. *(a)* $2\cos\theta = x + \dfrac{1}{x}$ ...(i)

On cubing both sides, we get $8\cos^3\theta = \left(x+\dfrac{1}{x}\right)^3$

$8\cos^3\theta = x^3 + \dfrac{1}{x^3} + 3x\cdot\dfrac{1}{x}\left(x+\dfrac{1}{x}\right)$

$8\cos^3\theta = x^3 + \dfrac{1}{x^3} + 3(2\cos\theta)$ [from Eq. (i)]

$\Rightarrow 8\cos^3\theta - 6\cos\theta = x^3 + \dfrac{1}{x^3}$

$\Rightarrow 2[4\cos^3\theta - 3\cos\theta] = x^3 + \dfrac{1}{x^3} \Rightarrow 2[\cos 3\theta] = x^3 + \dfrac{1}{x^3}$

88. *(a)* We know that, the value of $\cos\theta$ decreases in the interval $0 \le \theta \le \pi/2$

$\because \cos\theta \ge \dfrac{1}{2} \Rightarrow \cos\theta \ge \cos\dfrac{\pi}{3} \Rightarrow \theta \le \dfrac{\pi}{3}$

89. *(b)* Given, $\sin A = \dfrac{3}{5}$ and $\cos B = \dfrac{12}{13}$

$\therefore \tan A = \dfrac{\sin A}{\cos A} = \dfrac{\sin A}{\sqrt{1-\sin^2 A}} = \dfrac{3/5}{\sqrt{1-(3/5)^2}} = \dfrac{3/5}{4/5} = \dfrac{3}{4}$

and $\tan B = \dfrac{\sin B}{\cos B} = \dfrac{\sqrt{1-\cos^2 B}}{\cos B} = \dfrac{\sqrt{1-\dfrac{144}{169}}}{\dfrac{12}{13}} = \dfrac{5/13}{12/13} = \dfrac{5}{12}$

$$\frac{\tan A - \tan B}{1+\tan A\tan B} = \frac{\dfrac{3}{4}-\dfrac{5}{12}}{1+\dfrac{3}{4}\cdot\dfrac{5}{12}} = \frac{\dfrac{9-5}{12}}{\dfrac{48+15}{4\times 12}}$$

$$= \frac{\dfrac{1}{3}}{\dfrac{63}{48}} = \frac{1}{3}\times\frac{48}{63} = \frac{16}{63}$$

90. *(a)* Given, $B + C = 60° \Rightarrow B = 60° - C$

$\sin(120°-B) = \sin[120°-(60°-C)] = \sin(60°+C)$

$= \sin[180°-(60°+C)] = \sin(120°-C)$

91. *(a)* Given, $\dfrac{x\cdot\operatorname{cosec}^2 30°\sec^2 45°}{8\cos^2 45°\sin^2 60°} = \tan^2 60° - \tan^2 30°$

$\Rightarrow \dfrac{x\times(2)^2\times(\sqrt{2})^2}{8\times\left(\dfrac{1}{\sqrt{2}}\right)^2\times\left(\dfrac{\sqrt{3}}{2}\right)^2} = (\sqrt{3})^2 - \left(\dfrac{1}{\sqrt{3}}\right)^2$

$\Rightarrow \dfrac{x\times 4\times 2}{8\times\dfrac{1}{2}\times\dfrac{3}{4}} = 3 - \dfrac{1}{3} \Rightarrow \dfrac{8x}{3} = \dfrac{8}{3} \Rightarrow x = 1$

92. *(c)* $\dfrac{\sin x}{1+\cos x} = \dfrac{1-\cos x}{\sin x}$

The above identity is possible for all values of x except multiples of 180°. Since, for $x = 180°$, $\sin x = 0$ and $1+\cos x = 0$.

93. *(b)* Given, $\cos x = k\cos(x-2y) \Rightarrow \dfrac{1}{k} = \dfrac{\cos(x-2y)}{\cos x}$

Apply componendo and dividendo theorem

$$\frac{1-k}{1+k} = \frac{\cos(x-2y)-\cos x}{\cos(x-2y)+\cos x} = \frac{2\sin\dfrac{x-2y+x}{2}\sin\dfrac{x-x+2y}{2}}{2\cos\dfrac{x-2y+x}{2}\cos\dfrac{x-2y-x}{2}}$$

$$= \frac{2\sin(x-y)\sin y}{2\cos(x-y)\cos y} = \tan(x-y)\tan y$$

94. *(b)* $\dfrac{\sin(x+y)-2\sin x+\sin(x-y)}{\cos(x+y)-2\cos x+\cos(x-y)}$

$$= \frac{\sin(x+y)+\sin(x-y)-2\sin x}{\cos(x+y)+\cos(x-y)-2\cos x}$$

$$= \frac{2\sin\dfrac{x+y+x-y}{2}\cdot\cos\dfrac{x+y-x+y}{2}-2\sin x}{2\cos\dfrac{(x+y+x-y)}{2}\cdot\cos\dfrac{(x+y-x+y)}{2}-2\cos x}$$

$$= \frac{2\sin x\cos y-2\sin x}{2\cos x\cos y-2\cos x} = \frac{2\sin x(\cos y-1)}{2\cos x(\cos y-1)} = \frac{\sin x}{\cos x} = \tan x$$

95. *(c)* Given, $\sin x+\sin y = a \Rightarrow 2\sin\dfrac{x+y}{2}\cos\dfrac{x-y}{2} = a$ ...(i)

and $\cos x+\cos y = b \Rightarrow 2\cos\dfrac{x+y}{2}\cdot\cos\dfrac{(x-y)}{2} = b$ ...(ii)

On dividing Eq. (i) from Eq. (ii), we get

$$\therefore \quad \frac{2\sin\dfrac{x+y}{2}\cos\dfrac{x-y}{2}}{2\cos\dfrac{x+y}{2}\cos\dfrac{(x-y)}{2}} = \frac{a}{b} \Rightarrow \tan\frac{x+y}{2} = \frac{a}{b}$$

96. *(a)* Here, r = length of largest hand of clock

l = distance covered by largest hand of clock

Angle traced in 60 min = 360°

Angle traced in 20 min $= \dfrac{360°}{60}\times 20 = 120° \Rightarrow \theta = \dfrac{2\pi}{3}$

As, $\theta = \dfrac{l}{r} \Rightarrow l = r\theta$, $\Rightarrow l = 42\times\dfrac{2\pi}{3} = \dfrac{42\times2\times22}{3\times7} = 88$ cm

97. *(d)* Given, $2\sin x+15\cos^2 x = 7 \Rightarrow 2\sin x+15-15\sin^2 x = 7$

$\Rightarrow \quad 15\sin^2 x-2\sin x-8 = 0$

$\Rightarrow \quad 15\sin^2 x-12\sin x+10\sin x-8 = 0$

$\Rightarrow \quad 3\sin x(5\sin x-4)+2(5\sin x-4) = 0$

$\Rightarrow \quad (3\sin x+2)(5\sin x-4) = 0$

$\Rightarrow \quad \sin x = \dfrac{4}{5} \quad \left(\because \sin x \neq -\dfrac{2}{3}, 0° < x < 90°\right)$

$\therefore \quad \cos x = \sqrt{1-\sin^2 x} = \sqrt{1-\left(\dfrac{4}{5}\right)^2} = \sqrt{\left(\dfrac{3}{5}\right)^2} = \dfrac{3}{5}$

$\therefore \quad \tan x = \dfrac{\sin x}{\cos x} = \dfrac{4/5}{3/5} = \dfrac{4}{3}$

98. *(a)* In ΔABC, $A+B+C = 180°$

$\therefore \quad A+B = 180°-C \Rightarrow \tan(A+B) = \tan(180°-C)$

$\Rightarrow \quad \dfrac{\tan A+\tan B}{1-\tan A\tan B} = -\tan C$

$\Rightarrow \quad \tan A+\tan B = -\tan C+\tan A\tan B\tan C$

$\Rightarrow \quad \tan A+\tan B+\tan C = \tan A\tan B\tan C$

99. *(c)* Given, $x = a\sec\theta\cos\phi$, $y = b\sec\theta\sin\phi$ and $z = c\tan\theta$,

$$\frac{x^2}{a^2}+\frac{y^2}{b^2}-\frac{z^2}{c^2} = \frac{a^2\sec^2\theta\cos^2\phi}{a^2}+\frac{b^2\sec^2\theta\sin^2\phi}{b^2}-\frac{c^2\tan^2\theta}{c^2}$$

$$= \sec^2\theta[\cos^2\phi+\sin^2\phi]-\tan^2\theta = \sec^2\theta-\tan^2\theta = 1$$

100. *(a)* The squares of the tangents of the angles 30°, 45° and 60° are in G.P.

$\Rightarrow \tan^2 30°, \tan^2 45°, \tan^2 60°$ are in G.P. $\Rightarrow \left(\dfrac{1}{\sqrt{3}}\right)^2, (1)^2, (\sqrt{3})^2$

are in G.P. $\Rightarrow \dfrac{1}{3}, 1, 3$ are in G.P.

which is true as $1^2 = \dfrac{1}{3}\times 3 \Rightarrow 1 = 1$

101. *(d)* We know that, in a cyclic quadrilateral sum of opposite angle is 180°.

$\therefore \quad A+C = 180°$ ...(i)

and $\quad B+D = 180°$ ...(ii)

$\therefore \quad \cos A+\cos B+\cos C+\cos D$

$= \cos A+\cos B+\cos(180°-A)+\cos(180°-B)$

$= \cos A+\cos B-\cos A-\cos B = 0$

$[\therefore \cos(180°-\theta) = -\cos\theta]$

102. *(a)* We know that, for $0° < \theta < 90°$, there exist only one θ such that $\sin\theta = a$.

103. *(b)* $\because \sin(B+C-A) = 1 = \sin 90° \Rightarrow B+C-A = 90°$ ...(i)

$\because \quad \cos(C+A-B) = 1 = \cos 0°$

$\therefore \quad C+A-B = 0°$ ...(ii)

and $\tan(A+B-C) = 1 = \tan 45°$

$\therefore \quad A+B-C = 45°$ ...(iii)

On adding Eqs. (i), (ii) and (iii), we get

$A+B+C = 135°$ ...(iv)

On subtracting Eqs. (i), (ii), (iii) from Eq. (iv), we get

$2A = 45°, 2B = 135°, 2C = 90°$

$A = 22\dfrac{1°}{2}, B = 67\dfrac{1°}{2}, C = 45°$

104. *(c)* $\cos(180°+A)+\cos(180°+B)$

$+\cos(180°+C)+\cos(180°+D)$

$= -\cos A-\cos B-\cos C-\cos D$

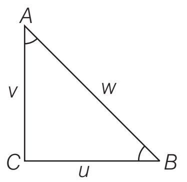
$= -\cos A-\cos B-\cos(180°-A)-\cos(180°-B)$

$(\because A+C = B+D = 180°)$

$= -\cos A-\cos B+\cos A+\cos B = 0$

105. *(d)* In ΔABC,

$$\tan A = \frac{BC}{AC} = \frac{u}{v} \text{ and } \tan B = \frac{v}{u}$$



Also, $u^2+v^2 = w^2$ (by pythagoras theorem) ...(i)

$\therefore \tan A+\tan B = \dfrac{u}{v}+\dfrac{v}{u} = \dfrac{u^2+v^2}{uv} = \dfrac{w^2}{uv}$ [from Eq. (i)]

106. *(a)* $\because$ In 24 h, Earth rotate about its own axis = 360°

In 1 h Earth rotate about its own axis

$= \dfrac{360°}{24} = 15°$

In 4 h Earth rotate about its own axis

$= 15° \times 4 = 60°$

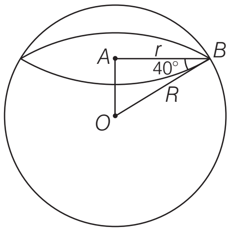
Since, in 60 min Earth rotate about its own axis = 15°

In 12 min Earth rotate about its own axis $= \dfrac{15° \times 12}{60°} = 3°$

$\therefore$ In 4 h 12 min Earth rotate about its own axis

$= 60° + 3° = 63°$

107. *(a)* In right angled ΔOAB,



$\cos 40° = \dfrac{AB}{OB} \Rightarrow \cos 40° = \dfrac{r}{R}$

$\Rightarrow \quad r = R \cos 40°$

So, the radius of the circle of latitude 40°S is $R \cos 40°$.

108. *(b)* Given, $\sin 3\theta = \cos(\theta - 2°)$

$\Rightarrow \quad \sin 3\theta = \sin[90° - (\theta - 2°)]$

$\Rightarrow \quad 3\theta = 90° - \theta + 2°$

$\Rightarrow \quad 4\theta = 92° \Rightarrow \theta = \dfrac{92°}{4} = 23°$

109. *(c)* I. RHS $= \cos^2\theta(1+\tan\theta)(1-\tan\theta)$

$= \cos^2\theta(1 - \tan^2\theta)$

$= \cos^2\theta\left(\dfrac{\cos^2\theta - \sin^2\theta}{\cos^2\theta}\right)$

$= \dfrac{\cos^2\theta - \sin^2\theta}{\cos^2\theta + \sin^2\theta} = \text{LHS}$

II. LHS $= \dfrac{1+\sin\theta}{1-\sin\theta} = \dfrac{(1+\sin\theta)^2}{1-\sin^2\theta}$

$= \left(\dfrac{1+\sin\theta}{\cos\theta}\right)^2$

$= (\sec\theta + \tan\theta)^2$

Hence, both statements are correct.

110. *(c)* Given, $A = \dfrac{\pi}{6}$ and $B = \dfrac{\pi}{3}$

I. LHS $\sin A + \sin B = \sin\dfrac{\pi}{6} + \sin\dfrac{\pi}{3}$

$= \dfrac{1}{2} + \dfrac{\sqrt{3}}{2} = \dfrac{1+\sqrt{3}}{2}$

RHS $\cos A + \cos B = \cos\dfrac{\pi}{6} + \dfrac{\cos\pi}{3}$

$= \dfrac{\sqrt{3}}{2} + \dfrac{1}{2} = \dfrac{\sqrt{3}+1}{2}$

$\Rightarrow \quad \sin A + \sin B = \cos A + \cos B$

II. LHS $\tan A + \tan B = \tan\dfrac{\pi}{6} + \tan\dfrac{\pi}{3}$

$= \dfrac{1}{\sqrt{3}} + \sqrt{3} = \dfrac{4}{\sqrt{3}}$

RHS $\cot A + \cot B = \cot\dfrac{\pi}{6} + \cot\dfrac{\pi}{3}$

$= \sqrt{3} + \dfrac{1}{\sqrt{3}} = \dfrac{4}{\sqrt{3}}$

$\Rightarrow \quad \tan A + \tan B = \cot A + \cot B$

Hence, both statements are true.

111. *(a)* $\because A + B + C + D = 360°$

$\therefore \quad A + B = 360° - (C + D)$

$\therefore \sin(A + B) = \sin[360° - (C + D)]$

$= -\sin(C + D)$

$\therefore \sin(A + B) + \sin(C + D) = 0$

Also, $\cos(A + B) = \cos[360° - (C + D)]$

$\cos(A + B) = \cos(C + D)$

Hence, only statement I is correct.

112. *(d)* I. $1° = \dfrac{\pi}{180}$ radian $= \dfrac{3.14}{180} = 0.017$

$= 0.02$ radians (approx)

which is less than 0.03 radians.

II. 1 radian $= \dfrac{180}{\pi}$ degree

$= \dfrac{180}{3.14} = 57.32$ degree

which is greater than 45°.

Hence, both statements are true.

113. *(d)* I. $\sin\theta \cdot \sin(60°+\theta) \cdot \sin(60° - \theta)$

$= \sin\theta[\sin^2 60° - \sin^2\theta]$

$= \sin\theta\left[\dfrac{3}{4} - \sin^2\theta\right]$

$= \dfrac{1}{4}[3\sin\theta - 4\sin^3\theta] = \dfrac{1}{4}\sin 3\theta$

II. $\cos\theta \cdot \sin(30°+\theta) \cdot \sin(30° - \theta)$

$= \cos\theta[\sin^2 30° - \sin^2\theta]$

$= \cos\theta\left[\dfrac{1}{4} - (1 - \cos^2\theta)\right]$

$= \dfrac{1}{4}[4\cos^3\theta - 3\cos\theta] = \dfrac{1}{4}\cos 3\theta$

III. $\sin\theta \cdot \cos(30°+\theta) \cdot \cos(30° - \theta)$

$= \sin\theta[\cos^2 30° - \sin^2\theta]$

$= \sin\theta\left[\dfrac{3}{4} - \sin^2\theta\right] = \dfrac{1}{4}\sin 3\theta$

Hence, all three statements are correct.

114. *(d)* In right angled ΔABC,

$A + B + C = 180°$

I. $\sin(A + B) = \sin(180° - C) = \sin C$

II. $\sin\left(\dfrac{A+B}{2}\right) = \sin\left(\dfrac{180° - C}{2}\right)$

$= \sin\left(90° - \dfrac{C}{2}\right) = \cos\dfrac{C}{2}$

III. $\tan\left(\dfrac{A+B-C}{2}\right)$

$= \tan\dfrac{(180° - C) - C}{2}$

$= \tan\dfrac{(180° - 2C)}{2}$

$= \tan(90° - C) = \cot C$

IV. $\tan\dfrac{(A - B - C)}{2} = \tan\dfrac{A - (B + C)}{2}$

$= \tan\dfrac{A - (180° - A)}{2}$

$= \tan[-(90° - A)]$

$= -\tan(90° - A) = -\cot A$

Hence, all options are correct.

115. *(b)* Given, $\sin(A + B) = 1$

and $\quad \sin(A - B) = 1/2$

$\therefore \quad A + B = \dfrac{\pi}{2} \quad \ldots\text{(i)}$

$\therefore \quad A - B = \dfrac{\pi}{6} \quad \ldots\text{(ii)}$

On adding Eqs. (i) and (ii), we get

$2A = \dfrac{2\pi}{3} \Rightarrow A = \dfrac{\pi}{3}$ and $B = \dfrac{\pi}{6}$

116. *(c)* Now, $\tan(A + 2B) \cdot \tan(2A + B)$

$= \tan\left(\dfrac{\pi}{3} + \dfrac{\pi}{3}\right) \cdot \tan\left(\dfrac{2\pi}{3} + \dfrac{\pi}{6}\right)$

$= \tan\left(\dfrac{2\pi}{3}\right) \cdot \tan\left(\dfrac{5\pi}{6}\right)$

$= \tan\left(\dfrac{\pi}{2} + \dfrac{\pi}{6}\right) \cdot \tan\left(\dfrac{\pi}{2} + \dfrac{\pi}{3}\right)$

$= \left[-\cot\dfrac{\pi}{6}\right]\left[-\cot\dfrac{\pi}{3}\right] = (-\sqrt{3})\left(\dfrac{-1}{\sqrt{3}}\right) = 1$

117. *(b)* Now, $\sin^2 A - \sin^2 B$

$= \sin^2(\pi/3) - \sin^2(\pi/6)$

$= \left(\dfrac{\sqrt{3}}{2}\right)^2 - \left(\dfrac{1}{2}\right)^2 = \dfrac{3}{4} - \dfrac{1}{4} = \dfrac{2}{4} = \dfrac{1}{2}$

118. *(c)* Since, $\tan A$, $\tan B$ are the roots of equation $3x^2 - 2\sqrt{3}x + 1 = 0$

$\therefore \quad \tan A + \tan B = \dfrac{2\sqrt{3}}{3}$

and $\quad \tan A \cdot \tan B = 1/3$

$\therefore \quad \tan(A + B) = \dfrac{\tan A + \tan B}{1 - \tan A \cdot \tan B}$

$= \dfrac{\frac{2\sqrt{3}}{3}}{1 - 1/3} = \sqrt{3} = \tan\dfrac{\pi}{3}$

$\therefore \quad A + B = \pi/3$

Now, $A + B + C = \pi$

$\Rightarrow \quad C = \pi - (A + B) = \pi - \dfrac{\pi}{3} = \dfrac{2\pi}{3}$

119. *(c)* Now, $\sin C + \cos C$
$= \sin(\pi - (A+B)) + \cos(\pi - (A+B))$
[as $A+B+C = \pi, C = \pi - (A+B)$]
$= \sin(A+B) - \cos(A+B)$
$= \sin(\pi/3) - \cos(\pi/3)$
$= \frac{\sqrt{3}}{2} - \frac{1}{2} = \frac{\sqrt{3}-1}{2}$

120. *(b)* $2\sin B = \sqrt{\frac{1+\tan^2 C}{\operatorname{cosec}^2 C}}$
$= \sqrt{\frac{1+\tan^2(2\pi/3)}{\operatorname{cosec}^2(2\pi/3)}}$
$= \sqrt{\frac{1+\tan^2(\pi-\pi/3)}{\operatorname{cosec}^2(\pi-\pi/3)}}$
$2\sin B = \sqrt{\frac{1+\tan^2 \pi/3}{\operatorname{cosec}^2 \pi/3}}$
$= \sqrt{\frac{1+3}{4/3}} = \sqrt{\frac{4}{4}\times 3} = \sqrt{3}$
$\sin B = \frac{\sqrt{3}}{2} = \sin\frac{\pi}{3}, B = \frac{\pi}{3}$

121. *(c)* Given that, $\sin^2 x + \cos^2 x - 1 = 0$
$\Rightarrow \sin^2 x + \cos^2 x = 1$
which is an identity of trigonometric ratio and always true for every real value of x.
So, the equation has infinite solutions.

122. *(c)* Given that, $3\sin x + 5\cos x = 5$
On squaring both sides, we get
$9\sin^2 x + 25\cos^2 x + 30\sin x\cos x = 25$
$\Rightarrow 9(1-\cos^2 x) + 25(1-\sin^2 x) + 30\sin x\cos x = 25$
$\Rightarrow 9 + 25 - \{9\cos^2 x + 25\sin^2 x - 30\sin x\cos x\} = 25$
$\Rightarrow 9 = (3\cos x - 5\sin x)^2$
$\Rightarrow 3\cos x - 5\sin x = 3$

123. *(c)* Given, $p = a\sin x + b\cos x$ and $q = a\cos x - b\sin x$
$\Rightarrow p^2 = a^2\sin^2 x + b^2\cos^2 x + 2ab\sin x\cos x$
and $q^2 = a^2\cos^2 x + b^2\sin^2 x - 2ab\sin x\cos x$
$\therefore p^2 + q^2 = a^2(\sin^2 x + \cos^2 x) + b^2(\cos^2 x + \sin^2 x)$
$= a^2 + b^2$

124. *(b)* Given, $\alpha + \beta = 90°$ ...(i)
$\therefore \sqrt{(\operatorname{cosec}\alpha \cdot \operatorname{cosec}\beta)}\left(\frac{\sin\alpha}{\sin\beta} + \frac{\cos\alpha}{\cos\beta}\right)^{-1/2}$
$= \frac{1}{(\sin\alpha\sin\beta)^{1/2}}\left(\frac{\sin\alpha\cos\beta + \cos\alpha\sin\beta}{\sin\beta\cos\beta}\right)^{-1/2}$
$= \frac{1}{(\sin\alpha\sin\beta)^{1/2}}\left\{\frac{\sin(\alpha+\beta)}{\sin\beta\cos\beta}\right\}^{-1/2}$
$= \frac{1}{(\sin\alpha\sin\beta)^{1/2}}\left\{\frac{\sin 90°}{\cos(90°-\alpha)\sin\beta}\right\}^{-1/2}$ [from Eq. (i)]
$= \frac{1}{(\sin\alpha\sin\beta)^{1/2}} \times (\sin\alpha\sin\beta)^{1/2} = 1$

125. *(c)* I. $\frac{\cot 30°+1}{\cot 30°-1} = 2(\cos 30° + 1)$
$\Rightarrow \frac{\sqrt{3}+1}{\sqrt{3}-1} = 2\left(\frac{\sqrt{3}}{2} + 1\right)$
$\Rightarrow \frac{\sqrt{3}+1}{\sqrt{3}-1} \times \frac{\sqrt{3}+1}{\sqrt{3}+1} = 2\left(\frac{\sqrt{3}+2}{2}\right)$
$\Rightarrow \frac{3+1+2\sqrt{3}}{3-1} = \sqrt{3} + 2$
$\Rightarrow \frac{4+2\sqrt{3}}{2} = \sqrt{3} + 2$
$\Rightarrow \frac{2(2+\sqrt{3})}{2} = \sqrt{3} + 2$
$\Rightarrow \sqrt{3} + 2 = \sqrt{3} + 2$
So, it is true.
II. $2\sin 45° \cos 45° - \tan 45° \cot 45° = 0$
$\Rightarrow 2 \times \left(\frac{1}{\sqrt{2}} \times \frac{1}{\sqrt{2}}\right) - 1 \times 1 = 0$
or $2 \times \frac{1}{2} - 1 \times 1 = 0$
$\Rightarrow 1 - 1 = 0$, It is true.
Hence, both statements I and II are true.

126. *(c)* I. We know that, $\sin^2\theta + \cos^2\theta = 1$
$\therefore \sin^2 1° + \cos^2 1° = 1$
II. We have
$\sec^2 33° - \cot^2 57°$
$= \operatorname{cosec}^2 37° - \tan^2 53°$
LHS $= \sec^2 33° - \cot^2 57°$
$= \sec^2(90°-57°) - \cot^2 57°$
$= \operatorname{cosec}^2 57° - \cot^2 57° = 1$
RHS $= \operatorname{cosec}^2 37° - \tan^2 53°$
$= \operatorname{cosec}^2 37° - \tan^2(90°-53°)$
$= \operatorname{cosec}^2 37° - \cot^2 37° = 1$
LHS = RHS
Hence, both statements are correct.

127. *(b)* I. Given that, $\sin x + \cos x = 2$
$\Rightarrow (\sin x + \cos x)^2 = 4$
$\Rightarrow (\sin^2 x + \cos^2 x) + 2\sin x\cos x = 4$
$\Rightarrow 1 + \sin 2x = 4 \Rightarrow \sin 2x = 3$
As maximum value of $\sin\theta$ is 1.
Therefore, no value of x satisfy above equation. Thus, statement I is false.
II. $\sin x - \cos x = 0$
$\Rightarrow \tan x = 1 = \tan\pi/4$
It x lies in first quadrant, then $\tan x = \tan\pi/4 \Rightarrow x = \pi/4$
Thus, statement II is true.

128. *(b)* Given that, $\sin\theta \cdot \cos\theta = \frac{\sqrt{3}}{4}$ ...(i)
$\therefore \sin^4\theta + \cos^4\theta$
$= (\sin^2\theta + \cos^2\theta)^2 - 2\sin^2\theta \cdot \cos^2\theta$
$= (1)^2 - 2(\sin\theta \cdot \cos\theta)^2$
$= 1 - 2\left(\frac{\sqrt{3}}{4}\right)^2 = 1 - 2 \cdot \frac{3}{16} = 1 - \frac{3}{8} = \frac{5}{8}$

129. *(d)* Given that, θ lies is in first quadrant and $\tan\theta = 3$
$\therefore \tan^2\theta = 9$
On adding both sides by 1, we get
$\Rightarrow 1 + \tan^2\theta = 10$
$\Rightarrow \sec^2\theta = 10 \Rightarrow \sec\theta = \sqrt{10}$
(since, θ lies in first quadrant)
$\Rightarrow \cos\theta = \frac{1}{\sqrt{10}}$ ...(i)
$\because \sin^2\theta = 1 - \cos^2\theta = 1 - \frac{1}{10} = \frac{9}{10}$
$\Rightarrow \sin\theta = \frac{3}{\sqrt{10}}$ ...(ii)
Now, $\sin\theta + \cos\theta = \frac{3}{\sqrt{10}} + \frac{1}{\sqrt{10}} = \frac{4}{\sqrt{10}}$

130. *(d)* If $0° < \theta < 90°$, then all the trigonometric ratios can be obtained when any one of the six ratios is given.

131. *(a)* $\sin A \cdot \cos A \cdot \tan A + \cos A \cdot \sin A \cdot \cot A$
$= \sin A \cdot \cos A \cdot \frac{\sin A}{\cos A} + \cos A \cdot \sin A \cdot \frac{\cos A}{\sin A}$
$= \sin^2 A + \cos^2 A$

132. *(a)* $\frac{\sin\theta}{1+\cos\theta} + \frac{1+\cos\theta}{\sin\theta}$
$= \frac{\sin^2\theta + (1+\cos\theta)^2}{\sin\theta(1+\cos\theta)}$
$= \frac{\sin^2\theta + 1 + \cos^2\theta + 2\cos\theta}{\sin\theta(1+\cos\theta)}$
$= \frac{\sin^2\theta + \cos^2\theta + 1 + 2\cos\theta}{\sin\theta(1+\cos\theta)}$
$= \frac{1+1+2\cos\theta}{\sin\theta(1+\cos\theta)}$
$= \frac{2(1+\cos\theta)}{\sin\theta(1+\cos\theta)} = 2\operatorname{cosec}\theta$

133. *(c)* $\sec^2 D - \tan^2 D = 1$
Since, it is a trigonometric identity.

134. *(b)* Given, $\cos A + \cos^2 A = 1$
$\Rightarrow \cos A = 1 - \cos^2 A = \sin^2 A$ ...(i)
$(\because \sin^2\theta + \cos^2\theta = 1)$
Now, $2(\sin^2 A + \sin^4 A)$
$= 2(\sin^2 A + \cos^2 A)$ [from Eq. (i)]
$= 2 \cdot (1) = 2$

135. *(c)* $(1-\tan A)^2+(1+\tan A)^2+(1-\cot A)^2+(1+\cot A)^2$

$= 2[1+\tan^2 A]+2[1+\cot^2 A]$

$[\because (a-b)^2+(a+b)^2=2(a^2+b^2)]$

$= 2\sec^2 A+2\operatorname{cosec}^2 A = 2\left(\frac{1}{\cos^2 A}+\frac{1}{\sin^2 A}\right)$

$= 2\left(\frac{\sin^2 A+\cos^2 A}{\sin^2 A\cdot\cos^2 A}\right)=\frac{2\cdot(1)}{\sin^2 A\cdot\cos^2 A}=2\sec^2 A\cdot\operatorname{cosec}^2 A$

136. *(d)* Given, $a^2=\frac{1+2\sin\theta\cdot\cos\theta}{1-2\sin\theta\cdot\cos\theta}$

$\Rightarrow a^2=\frac{(\sin^2\theta+\cos^2\theta)+2\sin\theta\cdot\cos\theta}{(\sin^2\theta+\cos^2\theta)-2\sin\theta\cdot\cos\theta}$

$\Rightarrow a^2=\frac{(\sin\theta+\cos\theta)^2}{(\sin\theta-\cos\theta)^2} \Rightarrow \frac{a}{1}=\frac{\sin\theta+\cos\theta}{\sin\theta-\cos\theta}$

$\Rightarrow \frac{a+1}{a-1}=\frac{(\sin\theta+\cos\theta)+(\sin\theta-\cos\theta)}{(\sin\theta+\cos\theta)-(\sin\theta-\cos\theta)}$

(applying componendo dividendo formula)

$\Rightarrow \frac{a+1}{a-1}=\frac{2\sin\theta}{2\cos\theta}=\tan\theta$

137. *(b)* Given, $\sin\theta=\frac{x^2-y^2}{x^2+y^2} \Rightarrow \cos^2\theta=1-\sin^2\theta=1-\left(\frac{x^2-y^2}{x^2+y^2}\right)^2$

$=\frac{(x^2+y^2)^2-(x^2-y^2)^2}{(x^2+y^2)^2}=\frac{4x^2y^2}{(x^2+y^2)^2}=\left(\frac{2xy}{x^2+y^2}\right)^2$

$\therefore \cos\theta=2xy/x^2+y^2$

138. *(b)* Let $f(\theta)=\sin\theta+\cos\theta$

Maximum and minimum value of $a\cos\theta+b\sin\theta$ is

$-\sqrt{a^2+b^2}\le a\cos\theta+b\sin\theta\le\sqrt{a^2+b^2}$

$\therefore -\sqrt{1+1}\le\cos\theta+\sin\theta\le\sqrt{1+1}$

$\Rightarrow -\sqrt{2}\le\cos\theta+\sin\theta\le\sqrt{2}$

for $0\le\theta\le 90°$, $1\le\cos\theta+\sin\theta\le\sqrt{2}$

Hence, statement I is false.

And let $g(\theta)=\tan\theta+\cot\theta=\tan\theta+\frac{1}{\tan\theta}$ $(\because \text{AM}\ge\text{GM})$

$\Rightarrow \frac{\tan\theta+\frac{1}{\tan\theta}}{2}\ge\left(\tan\theta\cdot\frac{1}{\tan\theta}\right)^{\frac{1}{2}}$

$\Rightarrow (\tan\theta+\cot\theta)\ge 2$

So, $(\tan\theta+\cot\theta)$ is always greater than 1.

Hence, statement II is true.

Solutions (Q. Nos. 139-141) Let the angle of A, B, C and D of a quadrilateral be x, $2x$, $4x$ and $5x$.

$\because x+2x+4x+5x=360°$

($\because$ sum of all interior in a quadrilateral is 360°)

$\Rightarrow 12x=360°, x=30°$

$\therefore \angle A=x=30°, \angle B=2x=60°$

$\angle C=4x=120°, \angle D=5x=150°$

139. *(a)* Now, $\cos(A+B)=\cos(30°+60°)=\cos 90°=0$

140. *(b)* Now, $\operatorname{cosec}(C-D+B)=\operatorname{cosec}(120°-150°+60°)$

$=\operatorname{cosec}(180°-150°)=\operatorname{cosec}30°=2$

141. *(d)* If $ABCD$ is a cyclic quadrilateral, then sum of opposite angles should be 180° but here

$\angle A+\angle C=30°+120°=150°\ne 180°$

and $\angle B+\angle D=60°+150°=210°\ne 180°$

So, statement I is incorrect.

Now, $\sin(B-A)=\cos(D-C)$

$\Rightarrow \sin(60°-30°)=\cos(150°-120°)$

$\Rightarrow \sin 30°=\cos 30° \Rightarrow \frac{1}{2}\ne\frac{\sqrt{3}}{2}$

So, statement II is also incorrect.

142. *(a)* Given, $\sin\theta+\cos\theta=\sqrt{3}$

On squaring both sides, we get $(\sin\theta+\cos\theta)^2=(\sqrt{3})^2$

$\Rightarrow \sin^2\theta+\cos^2\theta+2\sin\theta\cos\theta=3 \Rightarrow 1+2\sin\theta\cos\theta=3$

$\Rightarrow \sin\theta\cos\theta=\frac{3-1}{2}=1$ ...(i)

Now, $\tan\theta+\cot\theta=\frac{\sin\theta}{\cos\theta}+\frac{\cos\theta}{\sin\theta}=\frac{\sin^2\theta+\cos^2\theta}{\sin\theta\cos\theta}=\frac{1}{\sin\theta\cos\theta}$

From Eq. (i), $\tan\theta+\cot\theta=\frac{1}{1}=1$

143. *(a)* $\operatorname{cosec}(75°+\theta)-\sec(15°-\theta)=\operatorname{cosec}(75°+\theta)-\sec[90°-(75°+\theta)]$

$=\operatorname{cosec}(75°+\theta)-\operatorname{cosec}(75°+\theta)=0$

144. *(c)* Given, $5\sin\theta+12\cos\theta=13$

On squaring both sides, we get

$25\sin^2\theta+144\cos^2\theta+120\sin\theta\cos\theta=169$

$\Rightarrow 25(1-\cos^2\theta)+144(1-\sin^2\theta)+120\sin\theta\cos\theta=169$

$\Rightarrow 25-25\cos^2\theta+144-144\sin^2\theta+120\sin\theta\cos\theta=169$

$\Rightarrow 25\cos^2\theta+144\sin^2\theta-120\sin\theta\cos\theta=169-169$

$\Rightarrow (5\cos\theta-12\sin\theta)^2=0 \Rightarrow 5\cos\theta-12\sin\theta=0$

145. *(c)* Given, $\sin\theta-\cos\theta=0$

$\therefore \sin\theta=\cos\theta$

Since, $\sin\theta$ and $\cos\theta$ are equal for $\theta=45°$.

$\therefore \sin^4\theta+\cos^4\theta=(\sin 45°)^4+(\cos 45°)^4$

$=\left(\frac{1}{\sqrt{2}}\right)^4+\left(\frac{1}{\sqrt{2}}\right)^4=\frac{1}{4}+\frac{1}{4}=\frac{1+1}{4}=\frac{2}{4}=\frac{1}{2}$

146. *(c)* $\frac{\cos^2(45°+\theta)+\cos^2(45°-\theta)}{\tan(60°+\theta)\tan(30°-\theta)}$

$=\frac{\cos^2(45°+\theta)+\cos^2[90°-(45°+\theta)]}{\tan(60°+\theta)\cdot\tan[90°-(60°+\theta]}$

$=\frac{\cos^2(45°+\theta)+\sin^2(45°+\theta)}{\tan(60°+\theta)\cdot\cot(60°+\theta)}=\frac{1}{1}=1$

147. *(b)* Given, $\tan\theta+\sec\theta=m \Rightarrow \tan\theta=m-\sec\theta$

On squaring both sides, we get

$\Rightarrow \tan^2\theta=m^2+\sec^2\theta-2m\sec\theta$

$\Rightarrow \tan^2\theta=m^2+1+\tan^2\theta-2m\sec\theta$

$\Rightarrow 0=m^2+1-2m\sec\theta \Rightarrow \sec\theta=\frac{m^2+1}{2m}$